

Prediction and Reality in Australian Rooftop Solar

Forecasting Adoption and Learning from Where the Forecast Fails

Paper Outline — BDA 602 Module 4 Assignment

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2. Abstract

Australia reached high residential rooftop-solar penetration before almost any other country — roughly 4.5 million small-scale systems, about one in three homes — and consequently met the problems of success first: midday oversupply, negative wholesale prices, and, from July 2026, a policy offering households three hours of free daytime electricity to absorb surplus generation. Countries earlier on the same adoption curve are now forecasting their own transitions with comparable methods, so knowing where those methods break in a mature market is directly useful. This project asks how well household solar adoption can be predicted from the characteristics of a place, and, more importantly, what the prediction failures reveal. Using 25 years of monthly Clean Energy Regulator installation records for 2,811 Australian postcodes (854,310 postcode-month observations, April 2001 to July 2026) joined to Australian Bureau of Statistics Census demographics, we frame the supervised task as regression on postcode adoption rate, training on 2001–2015 and evaluating on 2016–2026 so that accuracy can be reported by prediction year. Ridge regression and random forest form the primary comparison, with a decision tree and gradient boosting (XGBoost) as additional models, all built in a reproducible Scikit-learn pipeline and tuned with grid, randomized, and Bayesian search under time-ordered validation.

The novelty lies in the unsupervised component. Rather than clustering postcodes by observed adoption, as prior Australian studies have done, we apply PCA and K-Means to each postcode's residual series — the direction, magnitude, and persistence of its prediction errors over time. Postcodes that consistently adopt far more or far less than their measured characteristics predict are where something outside the feature set was operating, and characterizing those groups demographically and geographically is the contribution. The expected impact is twofold: a temporally honest benchmark of how adoption forecasts degrade as a market matures, of direct use to grid and policy planners who build capacity on such forecasts, and a residual-based method for locating communities whose adoption behavior averages hide.

3. Introduction

3.1 Research question

- Primary question: how accurately can residential rooftop-solar adoption be predicted from the demographic, dwelling, and geographic characteristics of an Australian postcode, and how does that accuracy change as the prediction horizon extends beyond the training period?
- Secondary question: do the prediction errors form recurring, persistent patterns across postcodes, and what do the postcodes that systematically defy prediction have in common?
- The two questions are linked by design: the supervised model produces the residuals, and the unsupervised analysis is run on those residuals rather than on the raw data.

3.2 Background and motivation

- Australia is the world's leading residential rooftop-solar market by penetration; the Clean Energy Regulator records 4,501,887 small-scale solar installations since 2001, concentrated in Queensland (1.21M), New South Wales (1.18M), and Victoria (0.90M).
- Adoption arrived in distinct stages: near-zero activity until 2008, sharp peaks in 2011–2012 under generous state feed-in tariffs, a mid-2010s plateau, and renewed growth from 2017 as hardware costs fell. A model trained on one stage is not guaranteed to hold in the next.
- High penetration has created operational problems — midday oversupply exceeding state demand, negative wholesale prices, curtailment — and policy has shifted from encouraging adoption to managing its timing and location (e.g., the 2026 Solar Sharer Offer).
- Prior work (Zhang et al., 2023; Lan, Gou, & Liu, 2021; Best, Burke, & Nishitaten, 2019; CSIRO, 2024) has explained regional disparities cross-sectionally or forecast aggregate deployment; none we identified evaluates whether relationships learned in an earlier stage remain predictive in a substantially later one.

3.3 What is unique about this project

- Temporal holdout as a result, not a limitation: accuracy is reported by prediction year so that degradation with distance from the training period is itself a finding.
- Clustering the errors, not the data: prior studies cluster postcodes by observed penetration; we cluster by longitudinal residual profile, which identifies unusual adoption relative to expectation rather than high adoption per se.
- Documented handling of five data-quality characteristics in the CER data (leading-zero postcodes, schema drift between files, registration lag, privacy floor, no residential/commercial split), treated as design decisions rather than obstacles.
- Two explicit limitations carried through the paper: the analysis is ecological (postcode aggregates, not households) and descriptive/predictive, never causal.

3.4 Data and tools

- Data: Clean Energy Regulator small-scale installation postcode data (target); Australian Bureau of Statistics Census of Population and Housing (features and dwelling denominator).
- Tools: Python 3.10+, Pandas, NumPy, Matplotlib/Seaborn, Scikit-learn (Pipeline, ColumnTransformer, GridSearchCV, RandomizedSearchCV, PCA, KMeans), XGBoost, scikit-optimize (BayesSearchCV), MLflow, Git/GitHub.

4. Data Sources, Metadata, and Integration Strategy

4.1 Datasets

- **Dataset 1 — Small-scale Installation Postcode Data (Clean Energy Regulator, Australia).** Monthly counts and installed capacity of small-scale renewable installations by postcode, covering solar PV, solar water heating, heat pumps, and — from July 2025 — batteries. Distributed as CSV/XLSX in two files (2001–2010 and 2011–present), current as at 31 July 2026. Public, free, no registration. <https://cer.gov.au/markets/reports-and-data/small-scale-installation-postcode-data>
- **Dataset 2 — Census of Population and Housing (Australian Bureau of Statistics).** Postcode-level demographics: median household income, dwelling type, tenure, household composition, population density. Supplies the predictive features and the dwelling denominator that converts installation counts into adoption rates. Public and free through the ABS data portal. <https://www.abs.gov.au/census>

4.2 Metadata report — CER installation panel

Attribute	Value
Source files	SGU-Solar Installations 2001–2010 (2,809 × 118, wide) and 2011 to present (2,811 × 190, wide; two extra total columns)
Reshaped panel	854,310 rows = 2,811 postcodes × 304 months (April 2001 – July 2026), long format
Missing values	0 after reshaping
Reconciliation	Panel total of 4,501,887 installations matches the CER published figure exactly
Target distribution	Mean 5.27, median 0, SD 13.68, range 0–676; skewness 6.84, kurtosis 87.13
Sparsity	449,263 of 854,310 cells (52.6%) are zero; 669 postcodes have < 100 installations over 25 years
Concentration	Top 10% of postcodes account for 49.9% of all installations

4.3 Feature types

Field	Type	Notes
Postcode	Nominal categorical (stored as 4-character string)	Numeric storage implies a false ordering and silently drops leading zeros (0800 Darwin → 800)
Installation month	Datetime	Parsed from column headers during the melt
Installation count	Non-negative integer (target)	Converted to adoption rate via ABS dwelling denominator
Median household income	Continuous (ABS)	Examined jointly with density and dwelling type, not in isolation
Dwelling structure, tenure	Categorical (ABS)	Proportion detached / units / rented; captures rooftop access
Population density, median	Continuous (ABS)	Density identified as a key factor by Zhang et al.

age		(2023)
Privacy-floor flag	Binary (engineered)	Marks cells modified for privacy (< 10 installations)

4.4 Data-quality characteristics and handling decisions

- **Leading zeros.** 48 postcodes begin with 0. Read as text at import; otherwise the ABS join fails silently.
- **Schema drift.** The 2001–2010 file names monthly columns "Installations Quantity"; the 2011–present file uses "Installation Quantity". Reconciled during the melt; a JSON sidecar stores feature dtypes so type decisions survive across sessions.
- **Registration lag.** Installers have twelve months to register; national totals fall from ~40,790 (April 2026) to 17,708 (July 2026) because registration is incomplete, not because demand collapsed. The tail will be truncated or lag-adjusted.
- **Privacy floor.** Cells with fewer than ten installations are modified. Flagged as an indicator variable rather than treated as measurements.
- **No residential/commercial split.** A capacity threshold will be applied following the Australian PV Institute's approach, with sensitivity to the threshold reported.

4.5 Integration strategy

- Melt both CER files to long format, reconcile column names, parse month, and concatenate into one postcode-month panel.
- Join ABS dwelling counts and demographics on the string postcode to construct the adoption-rate target and the feature set.
- Census data is point-in-time (2021) while installations span 25 years; features are therefore static against a moving target, and the paper will distinguish retrospective explanation from genuine out-of-time forecasting where this matters (see Section 5.2).
- All preprocessing runs inside a Scikit-learn pipeline so that fitted transformations are learned from training folds only.

5. Methodology

5.1 Research design

- This project uses supervised and unsupervised machine learning to investigate rooftop solar adoption across Australian postcodes. The supervised component will estimate adoption using historical installation patterns and postcode characteristics. The unsupervised component will examine the resulting prediction errors to identify geographic areas exhibiting similar patterns of underprediction or overprediction.
- The analysis will use a chronological design, training supervised models on observations from 2001–2015 and evaluating their predictions on subsequent observations from 2016–2026, subject to excluding recent months affected by installation-registration delays.
- This approach allows us to examine not only how accurately rooftop solar adoption can be predicted, but also how prediction accuracy changes over time.

5.2 Data preparation and preprocessing

- The Clean Energy Regulator's installation records and Australian Bureau of Statistics demographic data will form the foundation of the modeling dataset.
- The existing installation data contains 854,310 postcode-month observations across 2,811 postcodes. The records cover April 2001 through July 2026.
- Preprocessing will include the following procedures:
 - **Postcode standardization:** Postcodes will be preserved as four-character strings to avoid losing leading zeros. This is necessary for matching installation records with demographic data.
 - **Data restructuring:** The two original Clean Energy Regulator files will be combined into a consistent longitudinal format. Differences in their column names will be reconciled, and installation months will be represented as datetime values.
 - **Data quality handling:** Recent observations affected by registration delays will be excluded or adjusted using a documented procedure. Privacy-modified installation counts will be flagged where identifiable, and sensitivity to those modifications will be considered. Residential-system filtering will be applied only if suitable system-size data are available.
 - **Demographic integration:** Postcode-level demographic variables, including household income, dwelling type, tenure, household composition, and population density, will be incorporated where appropriate data are available.
 - **Target construction:** The intended supervised target is postcode-level solar adoption rate, constructed using installation records and an appropriate dwelling denominator. The exact denominator and treatment of changes over time will be confirmed during data integration. If a defensible historical dwelling denominator is unavailable, installation counts will be modeled separately and clearly distinguished from adoption rates.
 - **Feature engineering:** Potential predictors include historical installation activity, calendar variables, geographic characteristics, and available demographic indicators. Historical installation features will be calculated using only information available before the prediction period.
- Numerical variables will be scaled where required, while categorical variables will be encoded using appropriate transformations. Missing demographic values will be handled using documented imputation or exclusion rules.
- A particular concern is that demographic information from the 2021 Census cannot automatically be treated as information available to a model forecasting from 2015. Where historically appropriate predictors are unavailable, the analysis will distinguish retrospective explanatory comparisons from genuine out-of-time forecasting.

5.3 Supervised learning methods

- The supervised task will be formulated as regression. Two principal algorithms will be evaluated.
- **Method 1: Ridge Regression**
 - Ridge regression will provide an interpretable baseline for estimating the relationship between postcode characteristics and solar adoption.
 - Its regularization parameter will be tuned to control the magnitude of model coefficients and reduce overfitting.

- The model will establish a reference point for determining whether more flexible machine-learning methods offer improvements in predictive performance.
- **Method 2: Random Forest Regression**
 - Random forest regression will be used to model nonlinear relationships and interactions among demographic, geographic, and historical installation features.
 - Unlike a linear model, a random forest can represent complex relationships without requiring every interaction to be specified manually.
 - The number of trees, maximum tree depth, and minimum number of samples required for splitting will be considered during hyperparameter tuning.
- The project may also compare a decision tree and gradient-boosted regression model, as proposed in the earlier planning summary, if time and computational resources permit. These will be additional supervised comparisons rather than substitutes for the two principal methods.

5.4 Unsupervised learning method

- **K-Means Clustering of Prediction Residuals**
- The unsupervised analysis will focus on the residuals produced by the supervised forecasting models.
- For each postcode and time period, the residual will be defined as: $\text{Residual} = \text{Actual adoption rate} - \text{Predicted adoption rate}$.
- A positive residual indicates that adoption exceeded the model's prediction. A negative residual indicates that adoption was lower than predicted.
- Residuals will be organized into time-ordered profiles for individual postcodes. These profiles will represent the direction, magnitude, and persistence of forecasting errors.
- Principal Component Analysis (PCA) will be considered to reduce dimensionality before clustering, while preserving enough information to distinguish persistent overprediction from persistent underprediction.
- K-Means will then group postcodes exhibiting similar residual patterns. The number of clusters will be selected using exploratory diagnostics, including silhouette scores and consideration of whether the resulting groups are interpretable.
- Hierarchical clustering may be used as an additional comparison.
- The resulting groups will be examined using available demographic and geographic information to identify characteristics associated with different forecasting-error patterns.
- Clustering will be conducted on out-of-sample residuals. Its purpose is retrospective error analysis, not to improve reported test-set performance using information from the test period.

5.5 Scikit-Learn pipeline and reproducibility

- The modeling workflow will be implemented primarily in Python using Scikit-Learn.
- A Pipeline and ColumnTransformer will organize preprocessing and supervised model estimation into reproducible workflows.
- Transformations requiring fitted parameters, such as scaling and imputation, will be learned from training data rather than from the complete dataset. The same fitted transformations will then be applied to validation and test observations.

- The planned workflow is: Raw installation data → Data cleaning → Demographic integration → Feature engineering → Chronological split → Preprocessing pipeline → Supervised modeling → Evaluation → Residual analysis → Clustering.
- Model configurations, random seeds, evaluation metrics, and generated artifacts will be recorded to support reproducibility. GitHub will be used for version control, and MLflow is planned for experiment tracking.

5.6 Evaluation and validation

- Supervised models will be evaluated using:
 - Mean Absolute Error (MAE), which measures the average absolute prediction error.
 - Root Mean Squared Error (RMSE), which gives greater weight to larger errors.
 - R-squared (R^2), which measures predictive fit relative to a mean-prediction baseline.
- These metrics will be reported on the held-out evaluation period, with performance also examined separately by prediction year.
- Time-ordered validation will be used during model development rather than randomly shuffling observations across years. Where feasible, additional geographic sensitivity checks will assess whether repeated observations from the same or nearby postcodes influence performance estimates.
- The final evaluation will preserve the distinction between model development and the held-out period. Models will not be selected or tuned using their final test-set performance.

6. Analysis Outline

6.1 Planned implementation

- The analysis will be implemented using Python, Pandas, NumPy, Scikit-Learn, and visualization libraries.
- The code will be organized into reusable components covering data preparation, feature engineering, model training, evaluation, and residual analysis.
- The following implementation stages are planned:
 1. Load the cleaned installation panel and available demographic data.
 2. Confirm postcode formatting, date ranges, missing values, and target construction.
 3. Develop historically appropriate predictive features.
 4. Establish chronological training, validation, and testing periods.
 5. Construct preprocessing and modeling pipelines.
 6. Train and compare the supervised algorithms.
 7. Tune model hyperparameters using time-ordered validation.
 8. Generate out-of-sample predictions and calculate evaluation metrics.
 9. Construct postcode-level residual profiles.
 10. Apply PCA and K-Means to the residual profiles.
 11. Interpret the resulting clusters using available demographic and geographic variables.

6.2 Exploratory analysis and planned figures

- The following figures will be prepared or incorporated into the analysis:
- **Figure 1. National solar installation trends, 2001–2026.** A time-series visualization showing the historical development of rooftop solar installations, including major changes in installation activity and the periods used for model training and testing.
- **Figure 2. Distribution of postcode-level installation activity.** A histogram or related distribution plot showing variation in installation counts or adoption rates, including the large number of low-activity observations.
- **Figure 3. Actual versus predicted adoption.** A comparison showing how closely each supervised model's predictions correspond to observed adoption during the held-out period.
- **Figure 4. Supervised model performance comparison.** A comparison of ridge regression and random forest using MAE, RMSE, and R^2 .
- **Figure 5. Prediction error over time.** A time-series visualization of model error by evaluation year to investigate changes in forecasting performance as the prediction horizon extends.
- **Figure 6. Residual patterns and clustering.** A visualization showing the major residual profiles identified by K-Means, supported by PCA plots or cluster-level time-series profiles where useful.
- **Figure 7. Geographic distribution of residual clusters.** A postcode-level map displaying the geographic distribution of residual groups, subject to availability of suitable postcode boundaries.

6.3 Model comparison

- The principal supervised comparison will examine ridge regression and random forest using the same target definition, training period, and held-out observations.
- Comparisons will consider overall predictive accuracy, error distribution, temporal stability, and interpretability.
- Where computationally feasible, additional models such as a decision tree or gradient boosting may be included.
- The analysis will examine whether the more flexible models capture nonlinear relationships that are not represented adequately by the linear baseline.
- However, the final comparison will be based on measured performance rather than assuming that a more complex algorithm must perform better.

6.4 Hyperparameter optimization

- Hyperparameter tuning will be performed using Scikit-Learn's GridSearchCV or RandomizedSearchCV with a suitable time-ordered validation strategy; scikit-optimize's BayesSearchCV will be used for the tree ensembles, where each fit is more expensive and the search space is wider.
- For ridge regression, the regularization parameter will be tuned.
- For random forest regression, candidate parameters will include the number of trees, maximum tree depth, minimum samples per split, and minimum samples per leaf.
- For K-Means, alternative numbers of clusters will be compared using clustering diagnostics and interpretability.
- Model tuning will be restricted to the development data. The final held-out period will be reserved for evaluation after model selection.

6.5 Residual analysis

- The residual analysis will investigate whether forecasting errors are randomly distributed or exhibit recurring patterns.
- Particular attention will be given to postcodes that experience persistent positive residuals, persistent negative residuals, or substantial changes in their prediction errors over time.
- The analysis will compare residual-cluster characteristics with available demographic and geographic variables.
- These comparisons will be interpreted as descriptive associations. They will not establish that any particular demographic characteristic or policy caused a postcode's forecasting error.

7. Projected Results

- The project is expected to produce a comparative assessment of supervised machine-learning methods for predicting Australian rooftop solar adoption.
- The ridge regression baseline will provide an interpretable reference for evaluating whether more flexible algorithms improve prediction accuracy. Random forest may capture nonlinear relationships and interactions among historical adoption patterns and postcode characteristics, although its relative performance will be established empirically.
- A second anticipated outcome is a clearer understanding of how forecasting performance changes over time. Evaluating the models across successive years will reveal whether predictive relationships remain stable or whether prediction errors increase, decrease, or change direction as the adoption market develops.
- The unsupervised analysis is expected to identify groups of postcodes with different forecasting-error profiles. Possible patterns include areas where adoption consistently exceeds predictions, areas where the model persistently overestimates adoption, and areas where errors change substantially over the evaluation period.
- These are hypotheses to investigate rather than findings already established by the data.
- Comparing the residual clusters with available demographic and geographic information may reveal associations that are obscured by national averages. The analysis may also identify limitations in the explanatory variables or modeling assumptions.
- The project's intended contribution is therefore twofold: a reproducible evaluation of rooftop solar adoption forecasting and a structured analysis of the places where those forecasts fail.
- The results may provide insights relevant to energy forecasting, infrastructure planning, and future research on geographically uneven technology adoption. Any practical recommendations will be based on the observed results and will distinguish predictive associations from causal explanations.

8. Discussion and Industry Applications

8.1 Interpreting the supervised results

- A widening error by prediction year would be consistent with Bass (1969): postcodes reach different points on their own diffusion curves at different times, so a relationship learned in the growth phase weakens as markets approach saturation. That is a substantive finding about forecasting maturity, not merely evidence of a weak model.

- If tree-based models outperform ridge regression, the improvement will be interpreted through the interactions the literature predicts — Lan, Gou, & Liu (2021) and Lan, Gou, & Lu (2021) show that higher income does not translate into higher adoption where density or apartment housing constrains rooftop access — rather than treated as a generic complexity benefit.
- Model trade-offs will be discussed explicitly: ridge is interpretable but linear; random forest and XGBoost capture interactions at the cost of interpretability and tuning effort; performance differences will be reported alongside run-to-run variation recorded in MLflow, not from a single run.

8.2 Interpreting the residual clusters

- A cluster with persistent positive residuals identifies postcodes where adoption ran ahead of what demographics predict; persistent negative residuals identify the reverse. Both are candidates for factors absent from the feature set — local policy, installer presence, community programs, or the payback period that CSIRO (2024) treats as the central driver and our features lack.
- The ecological-inference limitation applies throughout: clusters describe postcodes, not the households within them, and no cluster will be described as evidence that a demographic characteristic caused adoption.
- Results will be checked against the privacy-floor flag and the registration-lag truncation so that residual patterns are not artifacts of data handling.

8.3 Industry and policy applications

- Grid and market operators (e.g., AEMO, whose projections CSIRO prepares) build capacity plans on adoption forecasts; the size and direction of forecast error by region has direct planning consequences.
- Countries earlier on the adoption curve are using comparable postcode-level methods now; a documented account of where those methods break in a mature market transfers directly.
- The temporal-validation lesson is general for applied ML practice: a random split on spatially and temporally dependent data overstates performance (Roberts et al., 2017), and the honest evaluation is train on the past, test on the future.
- The residual-clustering approach is reusable for any geographically uneven technology adoption — EV charging, heat pumps, home batteries — where planners want to know which communities behave differently from what their demographics predict.

8.4 Limitations

- Features describe who lives in a postcode, not what a system costs them; payback period, electricity price, and local incentives are absent.
- Census demographics are point-in-time (2021) against a 25-year target; the analysis distinguishes retrospective explanation from genuine forecasting where this matters.
- CER data does not separate residential from commercial systems, and cells under ten installations are privacy-modified; both are handled by documented, sensitivity-tested rules rather than resolved.

9. Conclusion and Future Work

- The project delivers an end-to-end, reproducible ML pipeline on public data — two principal supervised models plus two additional ones, one unsupervised method, time-ordered tuning and evaluation, MLflow tracking, and a public GitHub repository — while shifting the analytical target from adoption alone to the behavior of prediction errors over time.
- Its contribution is longitudinal residual clustering: grouping postcodes by the magnitude, direction, and persistence of their forecast errors as evidence of factors missing from the measured feature set.
- Future work: (1) incorporate payback-period and electricity-price features to test whether residual clusters shrink; (2) extend the target to home batteries, which the CER data records from July 2025; (3) use multiple Census years with harmonized postcode boundaries so demographics move with the target; (4) add explicit spatial cross-validation (grouped by postcode or region) alongside the temporal holdout; (5) evaluate whether residual-cluster membership itself improves out-of-sample forecasts when used as a feature in a later period.

10. Author Biographies

Abraham Cho (acho8999@sdsu.edu) — Group Leader and Data Lead. Abraham is a graduate student in the Master of Science in Big Data Analytics program at San Diego State University and a secondary mathematics teacher with more than twenty years of classroom experience, holding an M.A. in Secondary Education. He is transitioning into data science and applied machine learning. His graduate coursework has included database management, web GIS, and a GIS study of urban heat, tree canopy, and walkability in San Diego's Convoy District. For this project he sources and documents the datasets, leads data quality assessment and preprocessing, owns the metadata report and the ABS demographic join, confirms that the temporal train/test split is reproducible and free of leakage, and submits all group deliverables.

Shashwat Bajaj (sbajaj5920@sdsu.edu) — Modeling Lead. Shashwat is a graduate student pursuing a Master of Science in Big Data Analytics at San Diego State University. His academic interests include machine learning, predictive modeling, and the application of data analytics to real-world challenges. His previous coursework has included the analysis of electric vehicle adoption and charging infrastructure in Southern California. As the Modeling Lead for this project, he focuses on developing the supervised and unsupervised machine-learning models, evaluating their performance, managing experiment tracking, and analyzing geographic patterns in rooftop solar adoption.

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